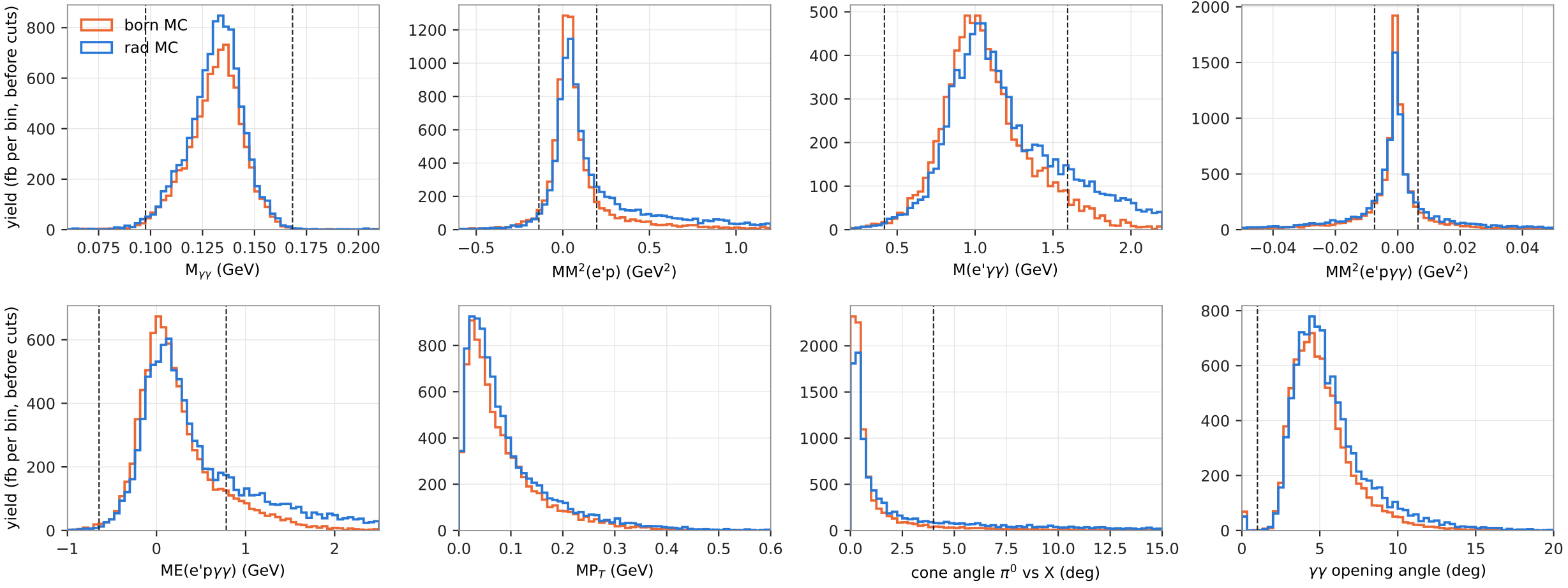
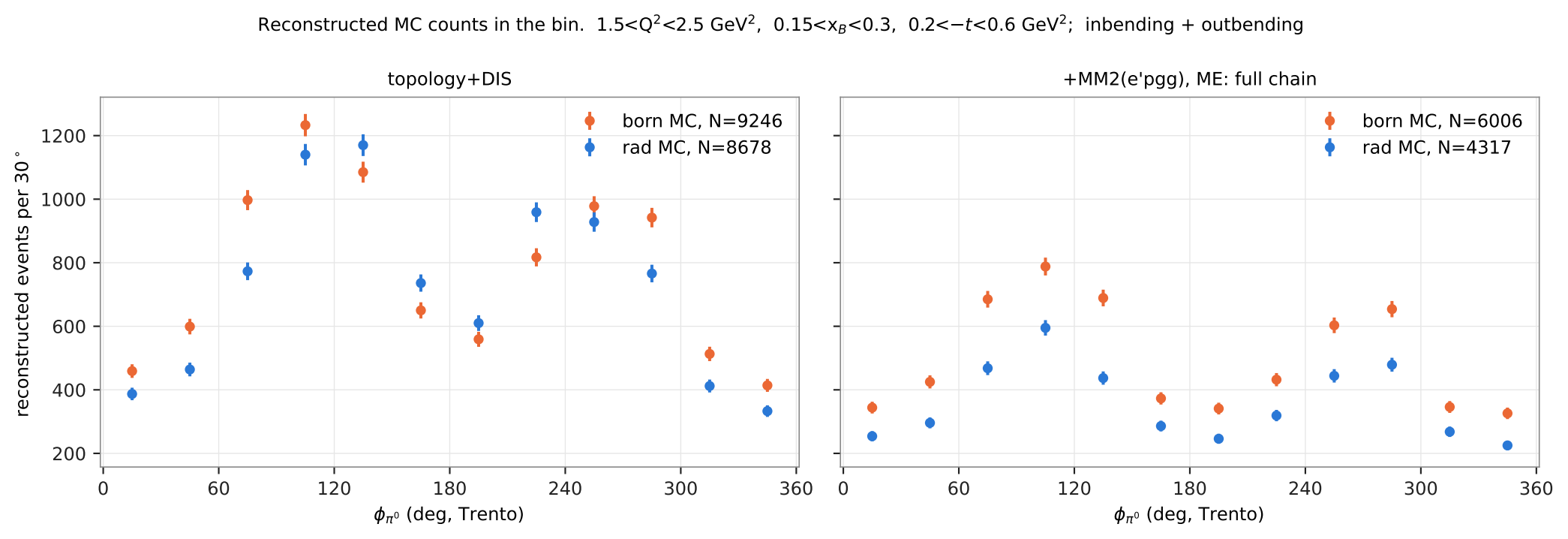
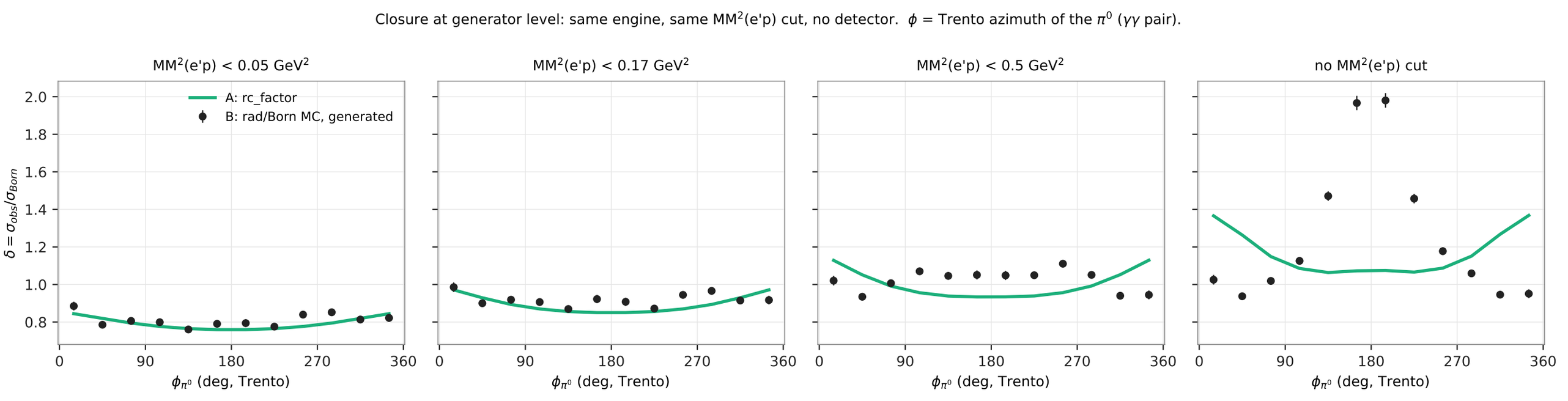
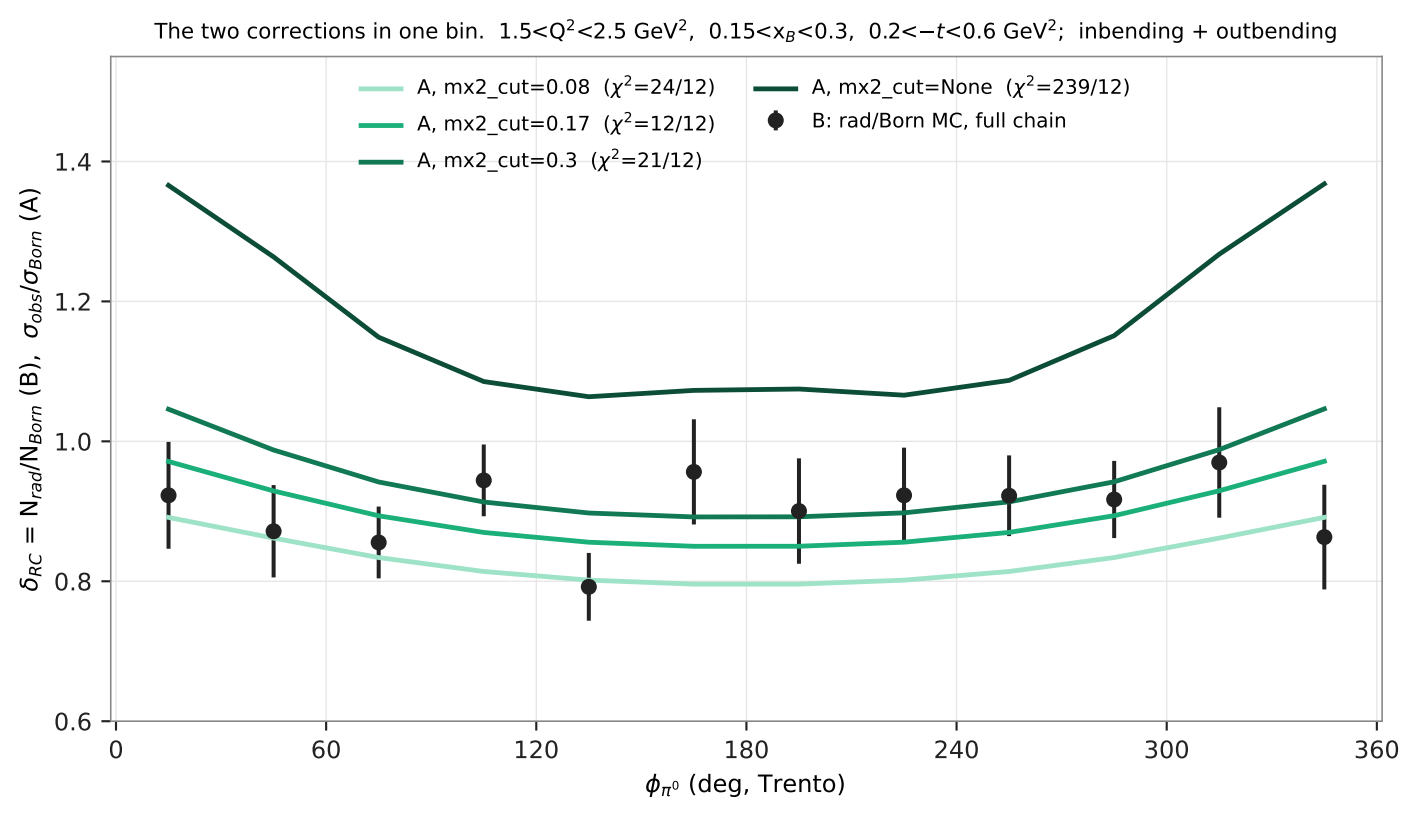
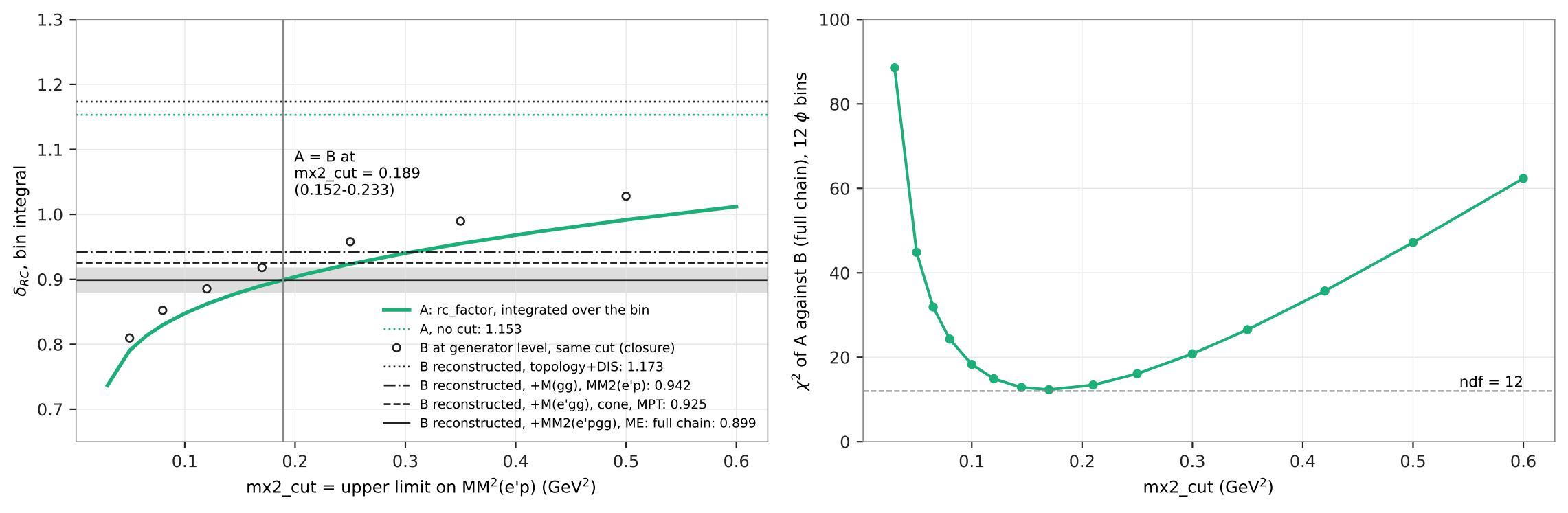
The MC in the bin after topology + DIS cuts, before exclusivity cuts. Dashed: the cut (3σ windows from the Born MC, outbending CD proton shown)
$1.5<Q^2<2.5$ GeV^2, $0.15<x_B<0.3$, $0.2<-t<0.6$ GeV^2; inbending + outbending




Reconstructed MC counts in the bin. 1.5<Q²<2.5 GeV², 0.15<x_B<0.3, 0.2<−t<0.6 GeV²; inbending + outbending
topology+DIS
+MM2(e'pgg), ME: full chain
born MC, N=9246
rad MC, N=8678
born MC, N=6006
rad MC, N=4317
reconstructed events per 30°
φ_π⁰ (deg, Trento)
φ_π⁰ (deg, Trento)
200
400
600
800
1000
1200
0
60
120
180
240
300
360


Closure at generator level: same engine, same MM²(e'p) cut, no detector. ϕ = Trento azimuth of the π^0 ($\gamma\gamma$ pair).
MM²(e'p) < 0.05 GeV²
MM²(e'p) < 0.17 GeV²
MM²(e'p) < 0.5 GeV²
no MM²(e'p) cut
A: rc_factor
B: rad/Born MC, generated
$\delta = \sigma_{obs}/\sigma_{Born}$
ϕ_{π^0} (deg, Trento)
0
90
180
270
360
0.8
1.0
1.2
1.4
1.6
1.8
2.0


The two corrections in one bin. 1.5<Q²<2.5 GeV², 0.15<x_B<0.3, 0.2<−t<0.6 GeV²; inbending + outbending
A, mx2_cut=0.08 (χ²=24/12)
A, mx2_cut=0.17 (χ²=12/12)
A, mx2_cut=0.3 (χ²=21/12)
A, mx2_cut=None (χ²=239/12)
B: rad/Born MC, full chain
δ_RC = N_rad/N_Born (B), σ_obs/σ_Born (A)
1.4
1.2
1.0
0.8
0.6
0
60
120
180
240
300
360
φ_π⁰ (deg, Trento)


A = B at
mx2_cut = 0.189
(0.152-0.233)
A: rc_factor, integrated over the bin
A, no cut: 1.153
B at generator level, same cut (closure)
B reconstructed, topology+DIS: 1.173
B reconstructed, +M(gg), MM2(e'p): 0.942
B reconstructed, +M(e'gg), cone, MPT: 0.925
B reconstructed, +MM2(e'pgg), ME: full chain: 0.899
δRC, bin integral
mx2_cut = upper limit on MM2(e'p) (GeV2)
χ2 of A against B (full chain), 12 ϕ bins
ndf = 12
mx2_cut (GeV2)